

Paper 1 (JINST) — proposed outline

For approval before writing. Draft 1, 8 October 2026

Working title

An algorithm-independent lower limit on combinatorial fake tracks in high-occupancy tracking detectors

(Alternatives: *How many random hit combinations look like a track? An exact estimate of irreducible fake tracks at high occupancy; Irreducible combinatorial fakes in tracking detectors: an analytic estimate and a Monte Carlo calibration at the 10^{-26} level.*)

Target: JINST regular article, about 20 pages, 7–8 figures. **Readers:** physicists who know track reconstruction at large collider experiments; no tutorial material on fits or Kalman filters.

The message in three sentences

1. When noise hits become dense enough compared with the detector resolution, some random combinations of hits fit a track as well as real tracks do; no algorithm that keeps real tracks at the same efficiency can reject them, because the information is not in the data.
2. Their expected number is exactly $E = P \times \prod n_i$ (P : probability that one random combination passes the χ^2 cut; n_i : noise hits per plane), and P , which can be 10^{-26} , is computed with an exact rescaling of the detector plus ordinary Monte Carlo.
3. E is therefore a lower limit on the fake rate of any algorithm; we show how it scales with hit density and resolution and apply it to a six-layer barrel tower with Muon Collider-like backgrounds.

Sections

1. Introduction (1.5 pp)

- High-occupancy tracking: HL-LHC, heavy ions, the Muon Collider beam-induced background. Fake rates are normally obtained by full simulation of one specific algorithm.
- The question asked here: at what density does tracking become impossible *for any algorithm*?

- Precursor: Casarsa, Jindariani, Ristori, JINST 20 (2025) P04030, Sec. 8.5 (inflated-resolution Monte Carlo scaled back with a product-of-resolutions law). This paper makes that argument exact and general.
- Related ideas outside tracking, cited and contrasted: random alignments of points (Kendall & Kendall; Broadbent), the “number of false alarms” of a contrario detection (Desolneux, Moisan, Morel), rare-event methods that inflate a scale and extrapolate (Bucher; Naess et al.). Accidental coincidences in counter systems and fake roads in associative-memory triggers as the familiar product-of-occupancies heuristics.
- Plan of the paper.

2. Definitions and the lower-limit statement (2 pp)

- Detector model: N planes, noise hits uniformly distributed with density ρ_i on plane i , measurement resolution σ (per coordinate, per plane).
- Track hypothesis and acceptance: least-squares fit, $\chi^2 < \text{cut}$, with the cut fixed by the efficiency for genuine tracks (one-sided 3σ : $P(\chi^2 > \text{cut}) = 1.35 \times 10^{-3}$).
- The counting identity $E = P \prod n_i$: exact for the mean by linearity of expectation, whatever the correlations between overlapping combinations.
- **The lower-limit statement** (wording as agreed): any algorithm that uses only the hit positions and keeps genuine tracks passing this χ^2 cut must also accept, on average, at least E pure-noise combinations. State the class of algorithms it covers; what can lower it (an algorithm that drops candidates through hit sharing or ambiguity resolution also loses real tracks) and what it does not count (partly-fake tracks, which only add to it).
- k of N planes: $E_{k\text{-of-}N} = \Sigma$ over subsets of k planes; bounds (largest subset $\leq$ distinct fakes $\leq$ sum).

3. The probability that one random combination passes (4 pp)

- Small- χ^2 power law $P(\chi^2 < c) \approx K c^{\text{ndof}/2}$: the residual vector lives in an ndof -dimensional space and its density near zero is constant; P is that density times the volume of an ellipsoid. Straight-line case reproduces the $(w/L)^{k-2}$ alignment law.
- Exact rescaling: shrinking the geometry by λ at fixed σ multiplies χ^2 by λ^2 exactly (residuals are linear in the data), so $P_{\text{true}}(c) = P_{\text{shrunk}}(c/\lambda^2)$; the rare event becomes common in the shrunk detector.
- Calibration: ordinary Monte Carlo on the shrunk detector; fit K on the plateau of $P/c^{\text{ndof}/2}$ with the exponent fixed; how to check the plateau; choice of λ (smallest plane $5\text{--}20\sigma$ wide).
- Consequences: $P \propto \sigma^{\text{ndof}}$ (resolution scaling); planes of different size and resolution through whitened coordinates.
- Relation to rare-event methods and to extreme-value theory: here the scaling is an exact symmetry and the exponent is known, so only K is fitted.

4. Validation (2.5 pp)

- The plateau of $P/c^{\text{ndof}/2}$ over several decades of c , straight line and helix (figure).

- The resolution law checked numerically (the $10^{\text{ndof}/N}$ ratio of crossing densities).
- **New:** an end-to-end test where fakes are frequent enough to count directly: generate noise-only events in a small detector, fit *all* combinations, count those passing the cut, and compare with $P \prod n_i$ (figure).

5. Tracks in a solenoidal field (3.5 pp)

- Tracks from the beam line: the fit splits into a bending view and a depth view, $\text{ndof} = 2N - 4$.
- Conservative quadratic model (linear, a superset of helices) and the exact helix fit in (b, κ) (Levenberg-Marquardt, several starting points; one sentence on why a single start underestimates fakes).
- Scale equivariance of the circle model: the rescaling stays exact provided curvature limits scale with λ ($\kappa \rightarrow \lambda\kappa$).
- Curvature acceptance: the accepted fraction grows linearly with κ_{max} and does not depend on σ .
- Comparison of the models; the $B = 0$ line as reference.

6. Worked example: a six-layer barrel tower (3 pp)

- Geometry and densities *inspired by* a Muon Collider tracker (citations): six layers at radii $\sim 160\text{--}1500$ mm, projective tower with 1 m^2 outermost plane, densities after background-suppression cuts of $10^{-4}\text{--}10^{-2}$ hits/ mm^2 , all rounded to two significant figures; $\sigma = 50\text{ }\mu\text{m}$; $B = 5\text{ T}$; $p_T > 10\text{ GeV}/c$.
- Results: P for the exact helix; E for 6 of 6 and for 5 of 6 (six subsets, compared).
- Headroom: E vs a common density factor ($\propto \mu^6$) and vs resolution ($\propto \sigma^8$), the factor at which one fake is reached.

7. Uncertainties and limitations (1.5 pp)

- Statistical uncertainty of K ; systematic from the plateau range; fit convergence.
- Assumptions: Gaussian resolution without tails, multiple scattering neglected (high- p_T tracks), projective idealization, noise uniform within a plane (local density is what enters).
- What would raise the true fake rate above the limit: partly-fake tracks, correlated noise (clusters, loopers).

8. Conclusions (0.5 pp)

Appendices

- A. Notation.
- B. Calibration constants (K , ndof , λ , statistics) for every configuration used.
- C. Code and data availability (git repository; archived version).
- Acknowledgements, including the AI disclosure statement required by IOP.

What is left out of the compendium

The history of the work and the “pitfalls” narratives (one sentence each where useful); the discussion on choosing λ (one rule of thumb); the angular-acceptance and impact-parameter cuts (one paragraph, or dropped); the first-order validity study (one remark); everything that refers to simulation data, which goes to paper 2.

New work needed before writing results

1. Recalibrate on the rounded geometry: exact helix for 6 of 6 and for each of the six 5-plane subsets (the slow runs; time to be estimated first).
2. Uncertainties: statistical (bootstrap) and plateau-range systematic for every K.
3. The end-to-end validation of Section 4 (new Monte Carlo, small detector).
4. All figures regenerated by scripts in docs/papers/jinst_fake_rate/.

Decisions for you

1. Title.
2. Keep the conservative quadratic model as a cross-check, or show only the exact helix (and the $B = 0$ line as reference)?
3. Angular-acceptance and impact-parameter cuts: one paragraph, or drop?
4. The end-to-end validation (Section 4): agree that it is worth the extra Monte Carlo?